

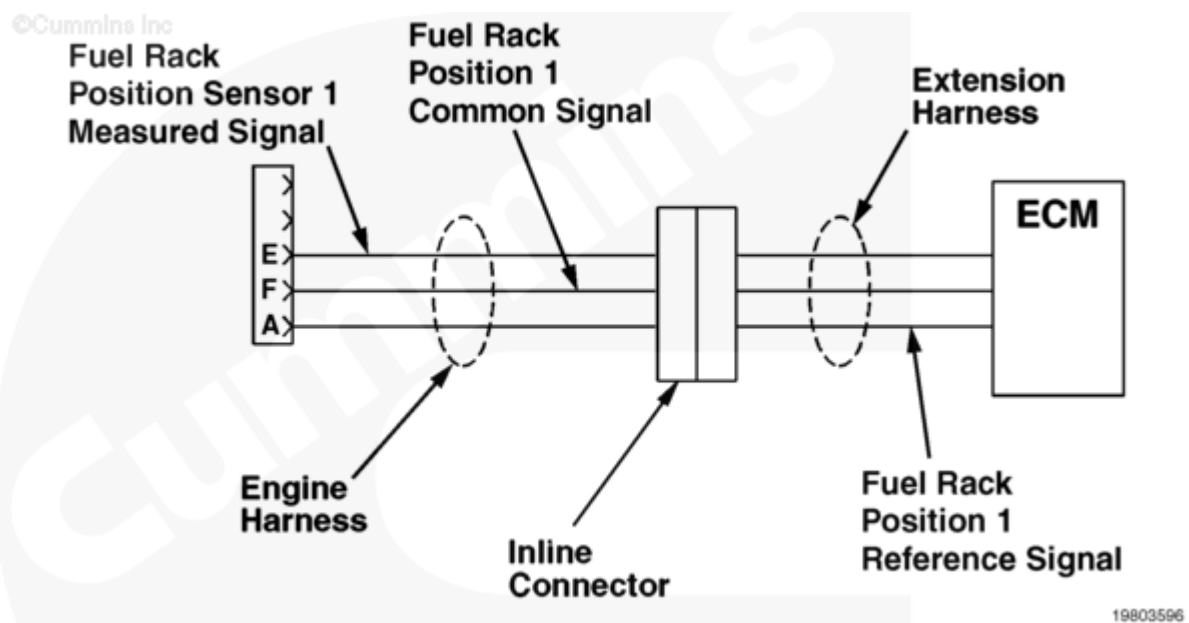
Fault Code: 2974

Fuel Rack Position 1 Circuit - Root Cause Not Known (Left Bank)

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 2974 PID(P): SPN: FMI: Lamp: Warning SRT:	Fuel rack position 1 circuit - root cause not known (left bank).	Engine can possibly run rough or overspeed and shutdown.



Fuel Rack Position Sensor 1 Circuit

Circuit Description

The ECM uses this rack position measured signal to determine if the fuel rack is at the commanded position. This is accomplished as follows:

The ECM sends out a fixed frequency and voltage signal on the reference signal line to the fuel pump. The ECM also sends out a fixed frequency but variable voltage signal that is 180 degrees out of phase from the reference signal from the measured signal line to the fuel pump. The

amplitude of the measured signal voltage is measured in the ECM and represents the rack position of the fuel pump.

Circuitry inside the fuel pump compares the difference in voltage between the reference signal input and the measured signal input to the pump and sends a signal back to the ECM that is the difference of these two signals. The difference voltage signal is output on the common signal line back to the ECM. The amplitude of the common signal is dependent on the position of the rack inside the fuel pump. Circuitry inside the ECM determines if the common signal is below a certain threshold. If it is, this means that the rack position of the fuel pump is at the commanded position. The voltage value of the measured signal line is then measured by the ECM and represents the rack position.

Component Location

The fuel rack position sensor 1 is located internal to the fuel pump on the left bank of the engine.

Shoptalk

The possible fail modes for this circuit are:

- Short to battery on the rack position measured signal, common signal, reference signal circuits
- Short pin to pin on the rack position measured signal, common signal, reference signal circuits
- Short to ground on the rack position measured signal, common signal, reference signal circuits
- Open circuit on the rack position measured signal, common signal, reference signal circuits
- Bad fuel pump
- Bad ECM.

This fault does **not** require engine speed to log a fault. The engine can overspeed and shut down. If Fault Code 234 (overspeed fault) and Fault Code 2974 are present, troubleshoot the Fault Code 2974 first. If you are monitoring the fuel rack position for the left bank, it will read zero millimeters (zero fueling).

If Fault Code 2974 and 169 are active, troubleshoot Fault Code 169 first.

Refer to Troubleshooting Fault Code t05-2974 (</qs3/pubsys2/xml/en/procedures/01/01-t05-2974.html>)

